

Natural Language Processing

Lecture 06: Machine Translation

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The course is delivered at ITMO University, Saint-Petersburg,
Bauman University, Moscow, IITU, Almaty

Content

- 1 Machine Translation (MT)
- 2 Machine translation evaluation
- 3 Statistical machine translation (SMT)
- 4 Neural machine translation (NMT) based on RNNs
- 5 Neural machine translation (NMT) with attentions

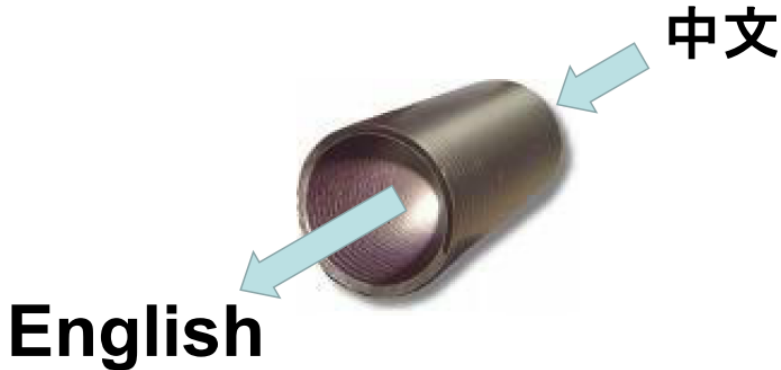
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Holy grails of NLP (Recap)

- **Accurate machine translation between human languages**
- Free conversation between humans and computers

Accurate machine translation (Recap)



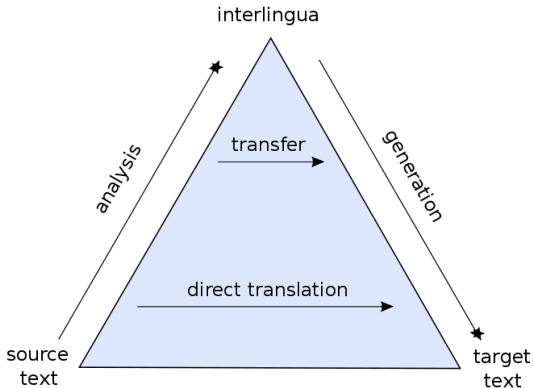
A brief history of MT

- 1940s-1950s: Early systems
- 1960s: ALPEC report (1966)
- 1970s: Operational systems: Systran, METEO
- 1980s: Eurotra project (EC-funded)
- Late 1980s–late1990s: the dawn of SMT (IBM), EBMT
- 2000s-early2010s: dominance of SMT
 - free on-line translation
- 2014-present: Neural MT

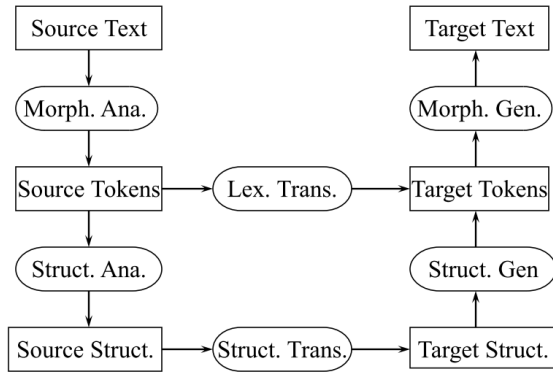
Different Approaches to MT

- Direct Translation
- Rule-based Machine Translation (RBMT)
- Memory-based Translation
- Example-based Machine Translation (EBMT)
- Statistical Machine Translation (SMT)
- Neural Machine Translation (NMT)

The Vauquois Pyramid

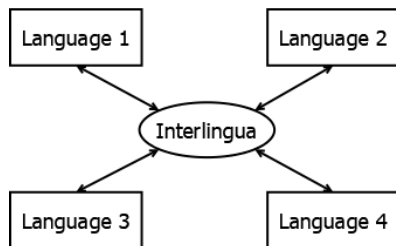


Transfer-based MT

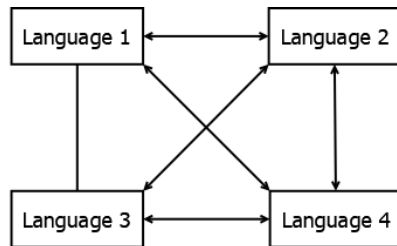


Transfer-based MT

Interlingua approach vs. transfer approach



Interlingua-based Approach



Transfer-based Approach

Multilingual MT

- Theoretically, interlingua-based approach is ideal because it requires much less components compared with transfer-based approach.
- However, using a human-defined interlingua (i.e. a specific knowledge expression) is practically too difficult.
- In business, a feasible way is to use a natural language (mostly English) as the interlingua, which is usually called a pivot language in this case.

Multilingual MT

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Multilingual MT

Makoto Nagao (Kyoto University):

When the pivot language (i.e. interlingua) is used, the results of the analytic stage must be in a form which can be utilized by all of the different languages into which translation is to take place. This level of subtlety is a practical impossibility. (Machine Translation, Oxford, 1989)

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Multilingual MT

Patel-Schneider (METAL system):

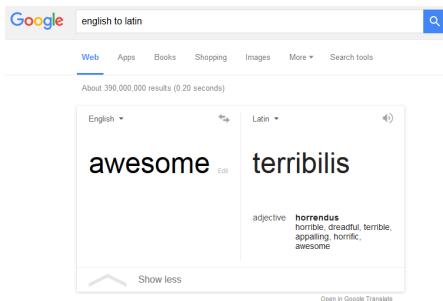
METAL employs a modified transfer approach rather than an interlingua. If a meta-language (an interlingua) were to be used for translation purposes, it would need to incorporate all possible features of many languages. That would not only be an endless task but probably a fruitless one as well. Such a system would soon become unmanageable and perhaps collapse under its own weight. (A four-valued semantics for terminological reasoning, Artificial Intelligence, 38, 1989)

Multilingual MT

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- In business, a feasible way is to use a natural language (mostly English) as the interlingua, which is usually called a pivot language in this case.

Applications of MT

Since neural machine translation become popular, the quality of MT systems is greatly improved, and MT is used in more and more scenarios.



Web Translator

Applications of MT

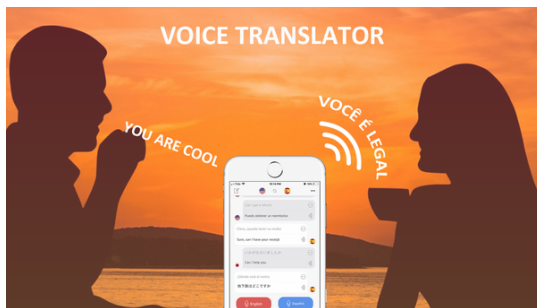
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Photo Translator

Applications of MT

Since neural machine translation become popular, the quality of MT systems is greatly improved, and MT is used in more and more scenarios.



Voice Translator

Applications of MT

Since neural machine translation become popular, the quality of MT systems is greatly improved, and MT is used in more and more scenarios.



Real-time Conference Translator

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Machine translation evaluation is not easy

- Machine translation evaluation is not easy because current translations for the same text may differ a lot literally.
- Human evaluation:
 - Reliable
 - Inconsistent
 - Expensive
 - Slow
- Automatic evaluation:
 - Unreliable
 - Consistent
 - Cheap
 - Fast

Goals for Automatic MT Evaluation

- **Meaningful:** score should give intuitive interpretation of translation quality.
- **Consistent:** repeated use of metric should give same results.
- **Correct:** metric must rank better systems higher.
- **Low cost:** reduce time and money spent on carrying out evaluation.
- **Tunable:** automatically optimise system performance towards metric.

Automatic MT Evaluation Metrics

- **Input:**

- output of machine translation systems.
- reference translations give by human translators.
- source text (optional).

- **Output:**

- a score which represents the similarity between the MT output and the human reference given the source sentence.

Automatic MT Evaluation Metrics

Plenty of automatic MT evaluation metrics are proposed by researchers, among which the most popular metrics include:

- Token-based metrics:
 - WER (Word Error Rate)
 - BLEU (BiLingual Evaluation Understudy)
 - TER (Translation Error Rate)
 - METEOR (Metric for Evaluation of Translation with Explicit ORdering)
 - chrF++ (Character F-score)
- Model-based metrics:
 - COMET (Cross-lingual Optimized Metric for Evaluation of Translation)

BLEU: Motivation

- N-gram precision between the machine translation and the reference
- Compute n-gram precisions for 1 to n (typically $n = 4$)
- A harmonic average of the n-gram precisions is calculated over different n 's
- Compute on the entire test set, rather than single sentences, to avoid 0 value for higher n-gram precisions

BLEU: Definition

$$\text{BLEU} = \underbrace{\min\left(1, \frac{\text{length_of_MT}}{\text{length_of_reference}}\right)}_{\text{Brevity Penalty}} \times \underbrace{\left(\prod_{i=1}^n \text{Precision}_i\right)^{\frac{1}{n}}}_{\text{N-gram Precision}}$$

$$\text{Precision}_i = \frac{\text{clipped_number_of_matched_}\{i\}\text{grams_in_MT}}{\text{number_of_total_}\{i\}\text{grams_in_MT}}$$

- Brevity Penalty is used to avoid that a very short MT get a high BLEU score.
- The number of matched $\{i\}$ grams is *clipped* to avoid that a single word in the reference is matched multiple times in MT.

Multiple References

To avoid translation variability, multiple references can be used:

- n-grams may match in any of the references
- closest reference length is used for brevity penalty

BLEU: An example

- **Candidate 1: the book is on the desk**

This example is extracted from Dr. Ming Zhou's slides on MT Evaluation

BLEU: An example

• **Candidate 1:** the book
is on the desk

• **Reference 1:** there is a book on the desk

• **Reference 2:** the book is on the table

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BLEU: An example

• **Candidate 1: the book is on the desk**

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unigram:

$$\text{Count}_{clip}(\text{the}) = 2$$

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$$\sum_{unigram \in C} \text{Count}(unigram) = 6$$

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$$p_1 = 1$$

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bigram:

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BLEU: An example

- **Candidate 1: the book is on the desk**

- **Reference 1: there is a book on the desk**
- **Reference 2: the book is on the table**

unigram:

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$$\text{Count}_{\text{clip}}(\text{the}, \text{desk}) = 1$$

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$$p_2 = 1$$

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trigram:

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bigram:

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$$\sum_{\text{trigram} \in C} \text{Count}(\text{trigram}) = 4$$

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$$\sum_{\text{unigram} \in C} \text{Count}(\text{unigram}) = 6$$

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$$\sum_{\text{bigram} \in C} \text{Count}(\text{bigram}) = 5$$

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trigram:

$$\text{Count}_{\text{clip}}(\text{the, book, is}) = 1$$

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$$p_3 = 1$$

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$$\sum_{\text{unigram} \in C} \text{Count}(\text{unigram}) = 6$$

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bigram:

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$$\text{Count}_{\text{clip}}(\text{is}, \text{on}) = 1$$

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$$\text{Count}_{\text{clip}}(\text{the}, \text{desk}) = 1$$

$$\sum_{\text{bigram} \in C} \text{Count}(\text{bigram}) = 5$$

$$p_2 = 1$$

trigram:

$$\text{Count}_{\text{clip}}(\text{the}, \text{book}, \text{is}) = 1$$

$$\text{Count}_{\text{clip}}(\text{book}, \text{is}, \text{on}) = 1$$

$$\text{Count}_{\text{clip}}(\text{is}, \text{on}, \text{the}) = 1$$

$$\text{Count}_{\text{clip}}(\text{on}, \text{the}, \text{desk}) = 1$$

$$\sum_{\text{trigram} \in C} \text{Count}(\text{trigram}) = 4$$

$$p_3 = 1$$

$$\left. \begin{array}{l} c = 6 \\ r = 6 \end{array} \right\} = e^{1 - \frac{r}{c}} = e^0 = 1 = BP$$

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$$p_3 = 1$$

$$\left. \begin{array}{l} c = 6 \\ r = 6 \end{array} \right\} = e^{1 - \frac{r}{c}} = e^0 = 1 = BP$$

$$\begin{aligned} \text{BLEU} &= BP \times (p_1 \times p_2 \times p_3)^{\frac{1}{3}} \\ &= 1 \times (1 \times 1 \times 1)^{\frac{1}{3}} = 1 \end{aligned}$$

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Content

3 Statistical machine translation (SMT)

- SMT: basic ideas
- Word-based Translation Models

Motivation

- In the time of rule-based MT, the translation knowledge was explicitly encoded by linguists, which is labor intensive, time consuming and inefficient.
- The basic idea of statistical machine translation is to learn a machine translation system directly from parallel translation data, without involving linguistics to encode translation knowledge.

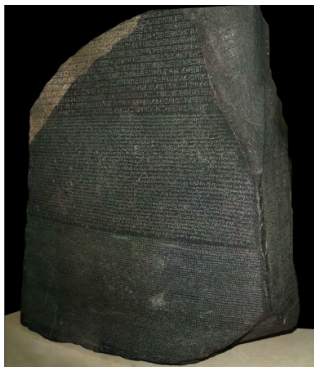
A brief history of SMT

- The first statistical machine translation was conducted by IBM researchers in late 1980s to early 1990s.
- The research of SMT was not very active in 1990s because the computing resources used by IBM researchers were higher than most researchers could have.
- In 1999, a group of researchers gathered in the JHU summer workshop and repeated IBM's early work successfully and release GIZA++, the implementation of the training algorithm of IBM models.
- After 2000 SMT became a dominate MT paradigm until it was replaced by NMT in recent years.

Parallel corpus

- A parallel corpus, or a bitext, is a collection of texts in one language and its corresponding translation in another language.
- Parallel corpora used for MT research are normally aligned at sentence level.
- Document-level MT needs parallel corpora aligned in both word level and document level.
- Many parallel corpora are aligned only in document level in their original form, so a technique called sentence alignment is developed to align the sentences in a document aligned corpus.

Rosette Stone



	The Rosetta Stone
Material	Granodiorite
Size	114.4 × 72.3 × 27.93 cm (45 x 28.5 x 11 in)
Writing	Ancient Egyptian hieroglyphs, Demotic script, and Greek script
Created	196 BC
Discovered	1799
Present	British Museum

Rosette stone is a good example that it is feasible to learn translation knowledge from a parallel corpus.

Statistical Translation Model

A statistical translation model is the probability that a target text e is the translation of a given source text f :

$$p(e|f), \text{ where } \sum_e p(e|f) = 1$$

Statistical Machine Translation

Given a statistical translation $p(e|f)$, the machine translation problem can be transferred to a search problem: to search a target sentence e which has the highest translation probability given f :

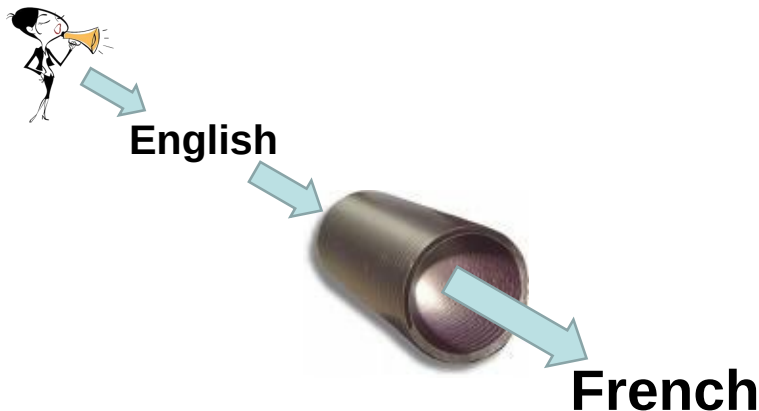
$$\hat{e} = \operatorname{argmax}_e p(e|f)$$

Noisy Channel Model

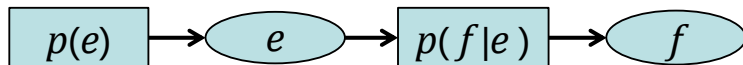
- Statistical machine translation with a single translation model does not work well.
- IBM researchers proposed a Noisy Channel Model for statistical machine translation, which includes an additional language model.

Peter F. Brown, John Cocke, Stephen A. Della Pietra, Vincent J. Della Pietra, Fredrick Jelinek, John D. Lafferty, Robert L. Mercer, Paul S. Roossin, A Statistical Approach to Machine Translation, Computational Linguistics, 1990

Noisy Channel Framework



Noisy Channel Framework

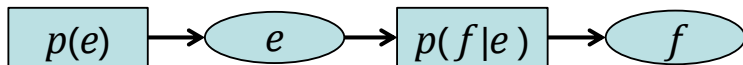


English



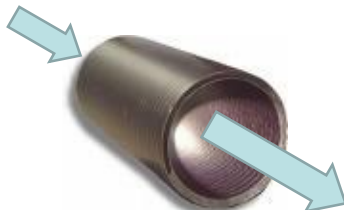
French

Noisy Channel Framework



English

$$p(f) = p(e)p(f|e)$$



French

Noisy Channel Framework

Applying Bayes' Rule, we have:

$$p(e|f) = \frac{p(e)p(f|e)}{p(f)}$$

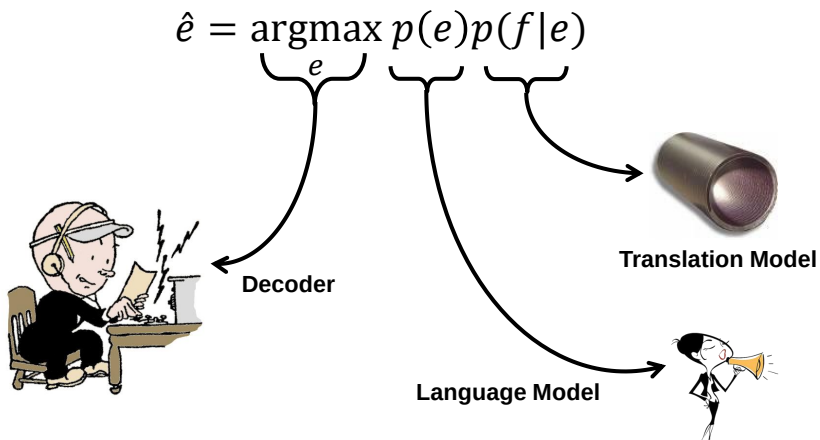
Thus:

$$\hat{e} = \operatorname{argmax}_e p(e|f) = \operatorname{argmax}_e p(e)p(f|e)$$

Noisy Channel Framework

$$\hat{e} = \operatorname{argmax}_e p(e)p(f|e)$$

Noisy Channel Framework



Noisy Channel Framework

- The *translation model* models how likely it is that f is a translations of e – **adequacy**.
- The *language model* models how likely it is that e is an acceptable sentence – **fluency**.
- The *decoder* searches for the most likely e .

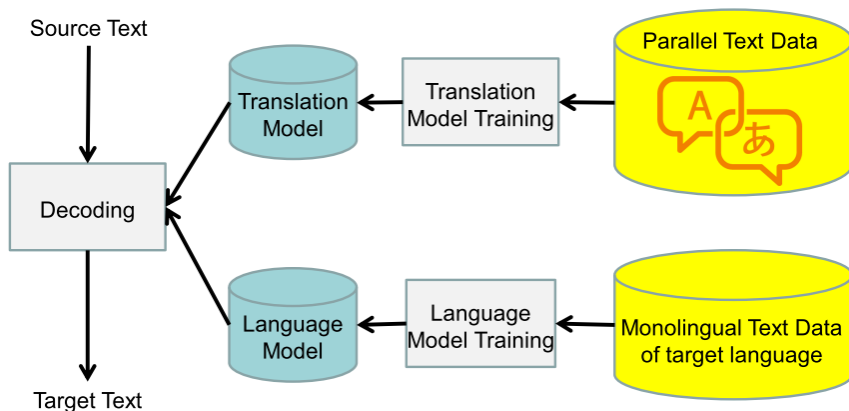
We have introduced language models in previous lectures, here we will mainly focus on translation models.

Noisy Channel Framework

- The *translation model* models how likely it is that f is a translations of e – **adequacy**.
- The *language model* models how likely it is that e is an acceptable sentence – **fluency**.
- The *decoder* searches for the most likely e .

We have introduced language models in previous lectures, here we will mainly focus on translation models.

SMT Workflow



Content

3 Statistical machine translation (SMT)

- SMT: basic ideas
- Word-based Translation Models

Categories of translation models

Various translation models have been proposed, which belong to different categories, according to the language units on which they are built up:

- Word-based models
 - IBM models 1-5
 - HMM models
- Phrase-based models
- Syntax-based models
 - Tree-to-string models
 - String-to-tree models
 - Tree-to-tree models
 - Dependency-based models

IBM Models

IBM researchers proposed 5 models with increased complexity:

- IBM Model 1: only consider lexical translation probabilities
- IBM Model 2: add a absolute reordering model
- IBM Model 3: add a fertility model
- IBM Model 4: add a relative reordering model
- IBM Model 5: improve previous model with more parameters

Lexical translation probabilities

German	English	Prob.
das	the	0.7
das	that	0.15
das	which	0.075
das	who	0.05
das	this	0.025
...	...	...

German	English	Prob.
Haus	house	0.8
Haus	building	0.16
Haus	home	0.02
Haus	household	0.015
Haus	shell	0.005
...	...	...

Word alignment

- To estimate the word translation probabilities, we need alignment between words in the parallel sentences

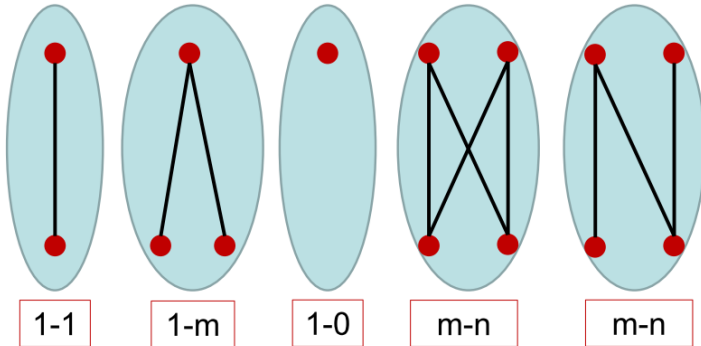
das Haus ist klein
 | | | |
the house is small

klein ist das Haus
 \ / \ /
the house is small

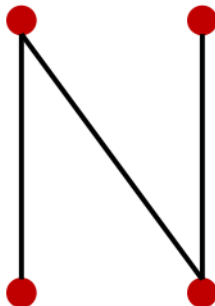
das Haus ist klitzeklein
 / | \ / \
the house is very small

das Haus ist klein
 | \ \
house is small

Word alignment patterns

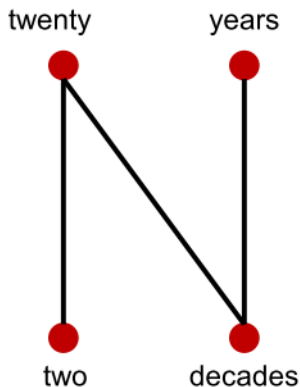


Word alignment patterns



Can you image a word alignment pattern like this?

Word alignment patterns



Can you image a word alignment pattern like this?

Learning lexical translation models

- We would like to estimate the lexical translation probabilities from a parallel corpus...
- but we do not have the alignments:
 - If we had the alignments, we could estimate the lexical translation probabilities.
 - If we had the probabilities, we could estimate the alignments.

Learning lexical translation models

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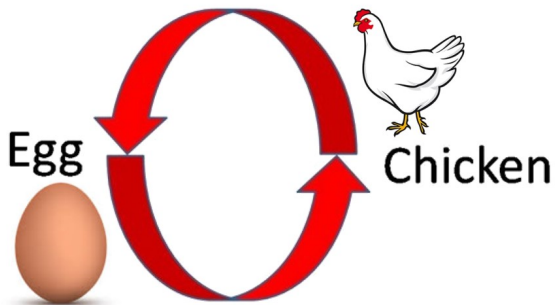
Learning lexical translation models

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Learning lexical translation models

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- but we do not have the alignments:
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 - If we had the probabilities, we could estimate the alignments.

A Paradox



EM Algorithm

- Incomplete data
 - If we had complete data, we could estimate model.
 - If we had the model, we could fill in the gaps in the data.
- Solution: **Expectation Maximization (EM)** Algorithm
 - Initialize model parameters. (e.g. uniform)
 - Assign probabilities to the missing data. (E-step)
 - Estimate model parameters from completed data. (M-step)
 - Iterate E-step and M-step until the model converges.

How does EM algorithm work?

EM Algorithm consists of two steps:

- 1 **Expectation-Step:** Apply model to the data
 - parts of the data are hidden (here: alignments)
 - using the model, assign probabilities of the hidden data to possible values (alignments)
- 2 **Maximization-Step:** Estimate new model from data
 - take assigned values as fact
 - collect counts (weighted by probabilities)
 - estimate new model from counts

Iterate the E-step and the M-step until convergence

Example

Consider a parallel corpus containing just two pairs:

blue house

house

maison bleu

maison

How many possible alignments in the first pair?

How many in the second pair?

Example

Consider a parallel corpus containing just two pairs:

blue house

house

maison bleu

maison

How many possible alignments in the first pair?

How many in the second pair?

We will simplify the example by ruling out many-to-one or zero-to-one alignments.

Example

Consider a parallel corpus containing just two pairs:

blue house

house

maison bleu

maison

How many possible alignments in the first pair? 2

How many in the second pair? 1

We will simplify the example by ruling out many-to-one or zero-to-one alignments.

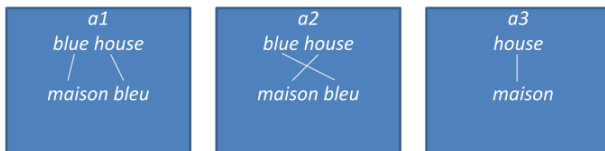
Step 1 (Initialisation)

Set parameter values uniformly.

- $t(\text{bleu}|\text{house}) = 1/2$
- $t(\text{maison}|\text{house}) = 1/2$
- $t(\text{bleu}|\text{blue}) = 1/2$
- $t(\text{maison}|\text{blue}) = 1/2$

Step 2 (Expectation)

Compute the probability of all alignments.



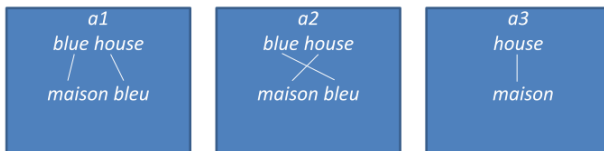
$$p(a1, \text{maison bleu} | \text{blue house}) = t(\text{maison} | \text{blue}) * t(\text{bleu} | \text{house}) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

$$p(a2, \text{maison bleu} | \text{blue house}) = t(\text{maison} | \text{house}) * t(\text{bleu} | \text{blue}) = \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$$

$$p(a3, \text{maison} | \text{house}) = t(\text{maison} | \text{house}) = \frac{1}{2}$$

Step 3 (Expectation)

Normalise for all alignments.



$$p(a1|\text{maison bleu}, \text{blue house}) = 1/4 \div 2/4 = 1/2$$

$$p(a2|\text{maison bleu}, \text{blue house}) = 1/4 \div 2/4 = 1/2$$

$$p(a3|\text{maison}, \text{house}) = 1/2 \div 1/2 = 1$$

Step 4 (Maximisation)

Collect fractional counts

- $tc(\text{bleu}|\text{house}) = 1/2$
- $tc(\text{maison}|\text{house}) = 1/2 + 1 = 3/2$
- $tc(\text{bleu}|\text{blue}) = 1/2$
- $tc(\text{maison}|\text{blue}) = 1/2$

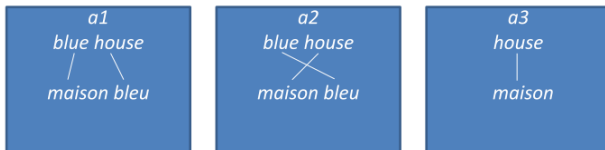
Step 5 (Maximisation)

Normalise fractional counts to yield revised parameter values

- $t(\text{bleu}|\text{house}) = 1/2 \div 2 = 1/4$
- $t(\text{maison}|\text{house}) = 3/2 \div 2 = 3/4$
- $t(\text{bleu}|\text{blue}) = 1/2 \div 1 = 1/2$
- $t(\text{maison}|\text{blue}) = 1/2 \div 1 = 1/2$

Repeat Step 2 (Expectation)

Compute the probability of all alignments.



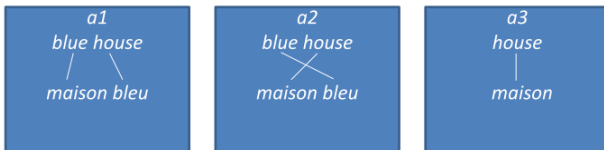
$$p(a1, \text{maison bleu} | \text{blue house}) = t(\text{maison} | \text{blue}) * t(\text{bleu} | \text{house}) = 1/2 * 1/4 = 1/8$$

$$p(a2, \text{maison bleu} | \text{blue house}) = t(\text{maison} | \text{house}) * t(\text{bleu} | \text{blue}) = 3/4 * 1/2 = 3/8$$

$$p(a3, \text{maison} | \text{house}) = t(\text{maison} | \text{house}) = 3/4$$

Repeat Step 3 (Expectation)

Normalise for all alignments.



$$p(a1|\text{maison bleu}, \text{blue house}) = 1/8 \div 4/8 = 1/4$$

$$p(a2|\text{maison bleu}, \text{blue house}) = 3/8 \div 4/8 = 3/4$$

$$p(a3|\text{maison}, \text{house}) = 3/4 \div 3/4 = 1$$

Repeat Step 4 (Maximisation)

Collect fractional counts

- $tc(\text{bleu}|\text{house}) = 1/4$
- $tc(\text{maison}|\text{house}) = 3/4 + 1 = 7/4$
- $tc(\text{bleu}|\text{blue}) = 3/4$
- $tc(\text{maison}|\text{blue}) = 1/4$

Repeat Step 5 (Maximisation)

Normalise fractional counts to yield revised parameter values

- $t(\text{bleu}|\text{house}) = 1/4 \div 2 = 1/8$
- $t(\text{maison}|\text{house}) = 7/4 \div 2 = 7/8$
- $t(\text{bleu}|\text{blue}) = 3/4 \div 1 = 3/4$
- $t(\text{maison}|\text{blue}) = 1/4 \div 1 = 1/4$

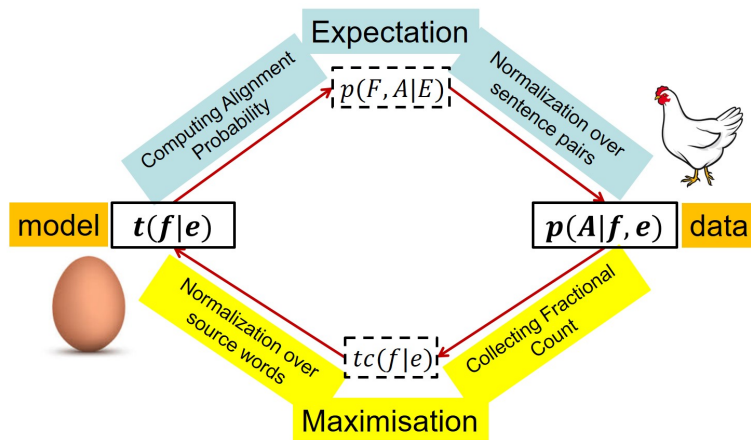
Convergence

Repeating steps 2, 3, 4 and 5 eventually yields:

- $t(\text{bleu}|\text{house}) = 0.0001$
- $t(\text{maison}|\text{house}) = 0.9999$
- $t(\text{bleu}|\text{blue}) = 0.9999$
- $t(\text{maison}|\text{blue}) = 0.0001$

It is proved that an EM algorithm is convergent.

EM Algorithm



Content

- 1 Machine Translation (MT)
- 2 Machine translation evaluation
- 3 Statistical machine translation (SMT)
- 4 Neural machine translation (NMT) based on RNNs**
- 5 Neural machine translation (NMT) with attentions

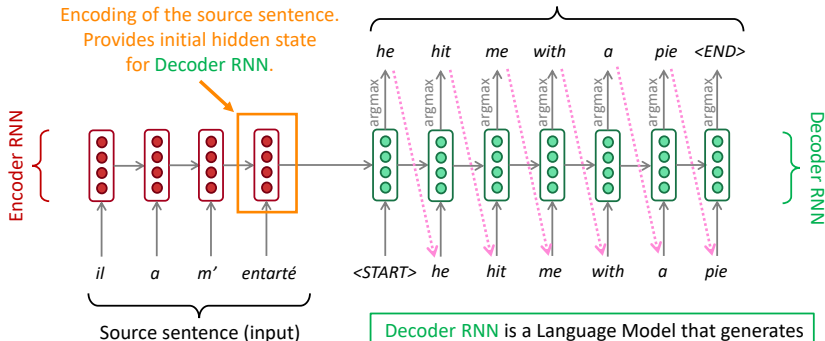
What is Neural Machine Translation?

- Neural Machine Translation (NMT) is a way to do Machine Translation with a *single neural network*
- The neural network architecture is called *sequence-to-sequence* (aka *seq2seq*) and it involves *two RNNs*.

Neural Machine Translation (NMT)

The sequence-to-sequence model

Encoding of the source sentence.
Provides initial hidden state
for Decoder RNN.



Encoder RNN produces
an **encoding** of the
source sentence.

Decoder RNN is a Language Model that generates
target sentence, *conditioned on encoding*.

Note: This diagram shows **test time** behavior:
decoder output is fed in> as next step's input

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Sequence-to-sequence is versatile!

- Sequence-to-sequence is useful for *more than just MT*
- Many NLP tasks can be phrased as sequence-to-sequence:
 - **Summarization** (long text → short text)
 - **Dialogue** (previous utterances → next utterance)
 - **Parsing** (input text → output parse as sequence)
 - **Code generation** (natural language → Python code)

Neural Machine Translation (NMT)

- The **sequence-to-sequence** model is an example of a **Conditional Language Model**.
 - **Language Model** because the decoder is predicting the next word of the target sentence y
 - **Conditional** because its predictions are *also* conditioned on the source sentence x

- NMT directly calculates $P(y|x)$:

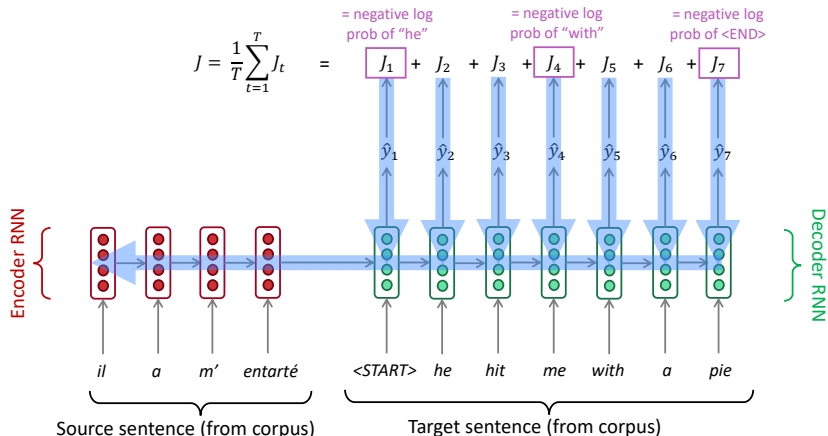
$$P(y|x) = P(y_1|x) P(y_2|y_1, x) P(y_3|y_1, y_2, x) \dots \underbrace{P(y_T|y_1, \dots, y_{T-1}, x)}$$

Probability of next target word, given target words so far and source sentence x

- **Question:** How to **train** a NMT system?
- **Answer:** Get a big parallel corpus...

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Training a Neural Machine Translation system



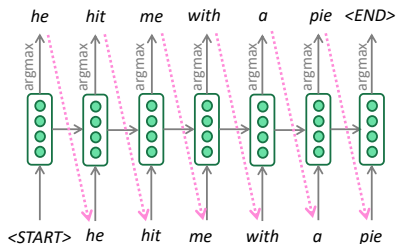
Seq2seq is optimized as a single system.
Backpropagation operates "end-to-end".

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Greedy decoding

- We saw how to generate (or “decode”) the target sentence by taking argmax on each step of the decoder



- This is **greedy decoding** (take most probable word on each step)
- Problems with this method?**

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Problems with greedy decoding

- Greedy decoding has no way to undo decisions!
 - Input: *il a m'entarté* (he hit me with a pie)
 - → *he* _____
 - → *he hit* _____
 - → *he hit a* _____ (whoops! no going back now...)
- How to fix this?

Exhaustive search decoding

- Ideally we want to find a (length T) translation y that maximizes

$$\begin{aligned} P(y|x) &= P(y_1|x) P(y_2|y_1, x) P(y_3|y_1, y_2, x) \dots, P(y_T|y_1, \dots, y_{T-1}, x) \\ &= \prod_{t=1}^T P(y_t|y_1, \dots, y_{t-1}, x) \end{aligned}$$

- We could try computing **all possible sequences** y
 - This means that on each step t of the decoder, we're tracking V^t possible partial translations, where V is vocab size
 - This $O(V^T)$ complexity is **far too expensive!**

Beam search decoding

- Core idea: On each step of decoder, keep track of the *k most probable* partial translations (which we call *hypotheses*)
 - *k* is the *beam size* (in practice around 5 to 10)
- A hypothesis $y_1, \dots, y_t$ has a *score* which is its log probability:

$$\text{score}(y_1, \dots, y_t) = \log P_{\text{LM}}(y_1, \dots, y_t | x) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

- Scores are all negative, and higher score is better
 - We search for high-scoring hypotheses, tracking top *k* on each step
- Beam search is *not guaranteed* to find optimal solution
- But *much more efficient* than exhaustive search!

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$

<START>

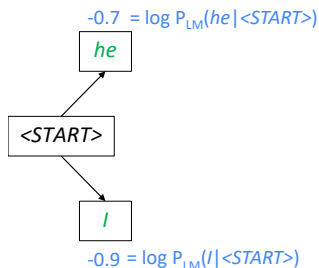
Calculate prob
dist of next word

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



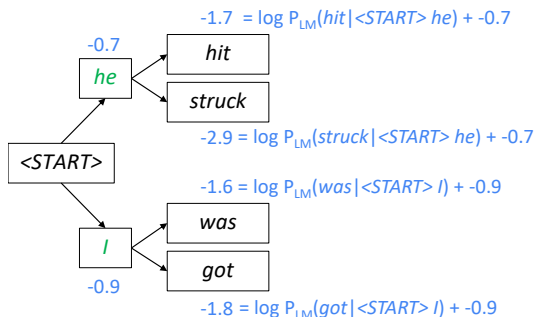
Take top k words
and compute scores

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



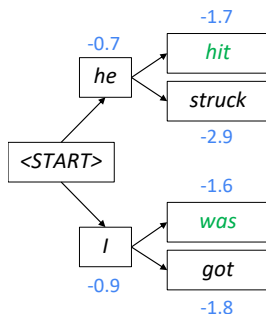
For each of the k hypotheses, find top k next words and calculate scores

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



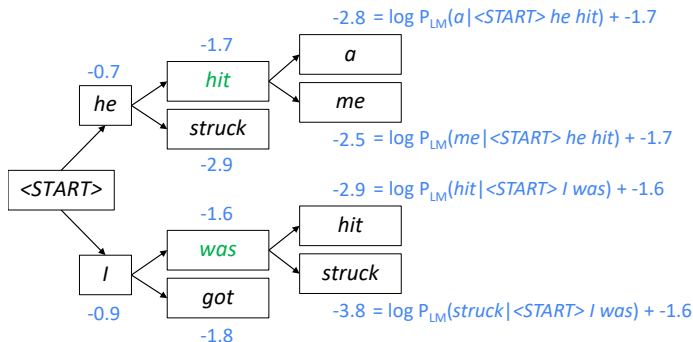
Of these k^2 hypotheses,
just keep k with highest scores

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



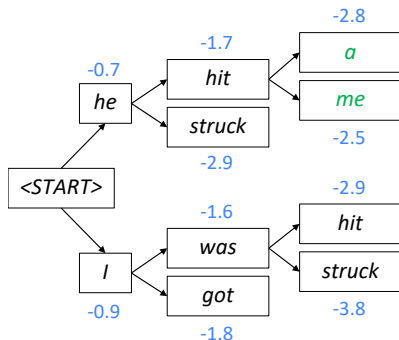
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Beam search decoding: example

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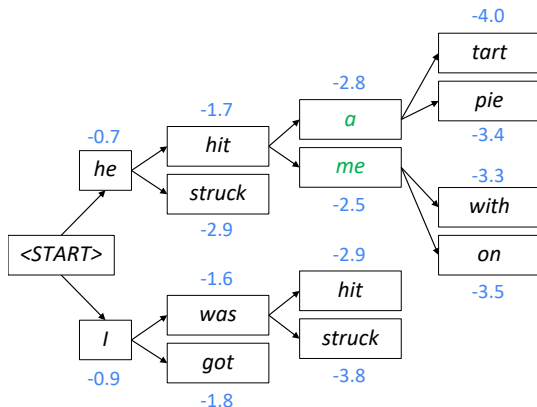
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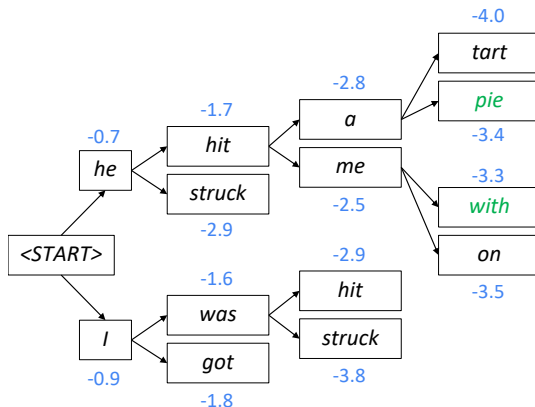
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Beam search decoding: example

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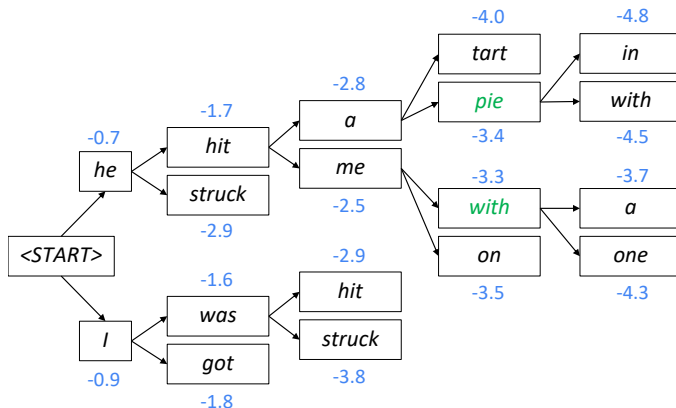
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39

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Beam search decoding: example

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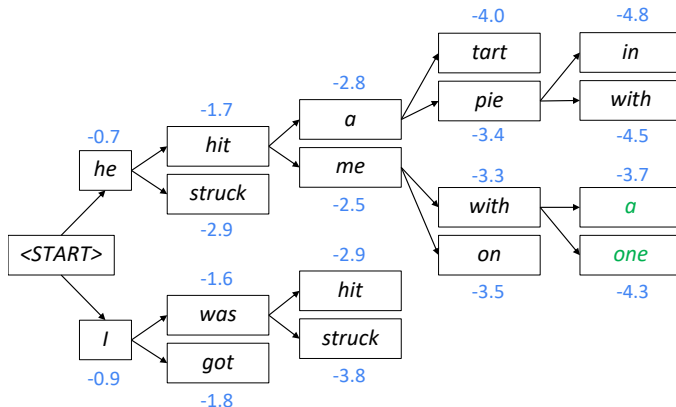
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Beam search decoding: example

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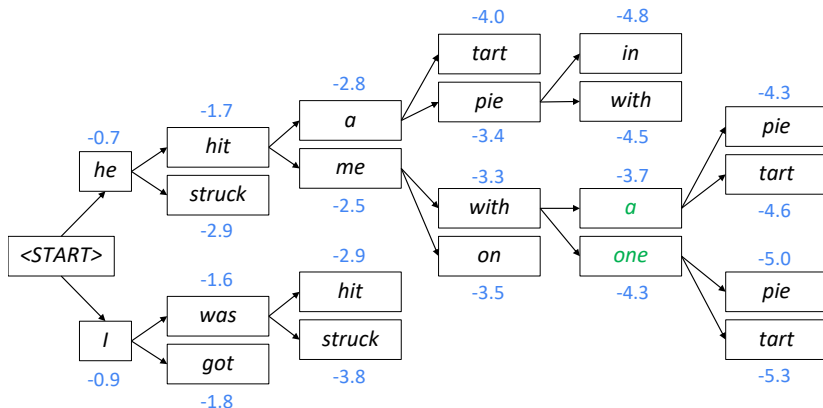
Of these k^2 hypotheses,
just keep k with highest scores

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



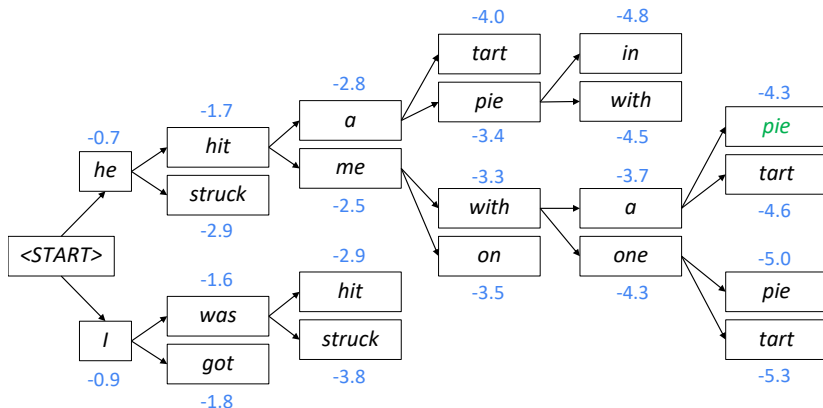
For each of the k hypotheses, find top k next words and calculate scores

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



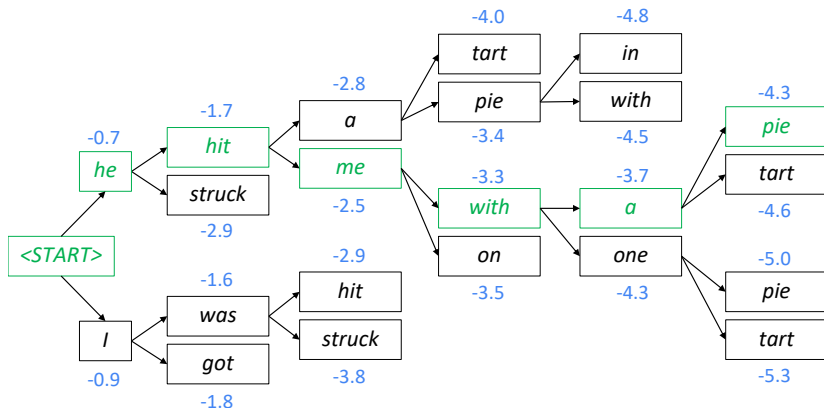
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This is the top-scoring hypothesis!

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Beam search decoding: example

Beam size = $k = 2$. Blue numbers = $\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$



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Backtrack to obtain the full hypothesis

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Beam search decoding: stopping criterion

- In **greedy decoding**, usually we decode until the model produces a **<END> token**
 - For example: *<START> he hit me with a pie <END>*
- In **beam search decoding**, different hypotheses may produce **<END> tokens on different timesteps**
 - When a hypothesis produces **<END>**, that hypothesis is **complete**.
 - **Place it aside** and continue exploring other hypotheses via beam search.
- Usually we continue beam search until:
 - We reach timestep T (where T is some pre-defined cutoff), or
 - We have at least n completed hypotheses (where n is pre-defined cutoff)

Beam search decoding: finishing up

- We have our list of completed hypotheses.
- How to select top one with highest score?
- Each hypothesis $y_1, \dots, y_t$ on our list has a score

$$\text{score}(y_1, \dots, y_t) = \log P_{\text{LM}}(y_1, \dots, y_t | x) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

- Problem with this: longer hypotheses have lower scores
- Fix: Normalize by length. Use this to select top one instead:

$$\frac{1}{t} \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

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Advantages of NMT

Compared to SMT, NMT has many **advantages**:

- Better **performance**
 - More **fluent**
 - Better use of **context**
 - Better use of **phrase similarities**
- A **single neural network** to be optimized end-to-end
 - No subcomponents to be individually optimized
- Requires much **less human engineering effort**
 - No feature engineering
 - Same method for all language pairs

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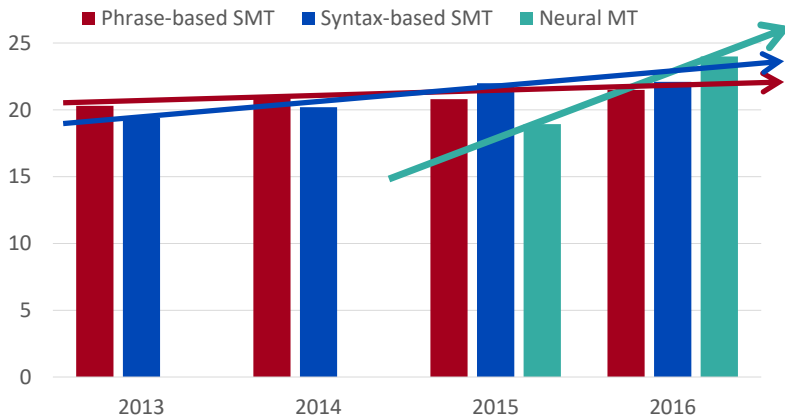
Disadvantages of NMT?

Compared to SMT:

- NMT is **less interpretable**
 - Hard to debug
- NMT is **difficult to control**
 - For example, can't easily specify rules or guidelines for translation
 - Safety concerns!

MT progress over time

[Edinburgh En-De WMT newstest2013 Cased BLEU; NMT 2015 from U. Montréal]



Source: http://www.meta-net.eu/events/meta-forum-2016/slides/09_sennrich.pdf

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NMT: the biggest success story of NLP Deep Learning

Neural Machine Translation went from a **fringe research activity** in **2014** to the **leading standard method** in **2016**

- **2014**: First seq2seq paper published
- **2016**: Google Translate switches from SMT to NMT
- **This is amazing!**
 - **SMT** systems, built by **hundreds** of engineers over many **years**, outperformed by NMT systems trained by a **handful** of engineers in a few **months**

So is Machine Translation solved?

- **Nope!**
- Many difficulties remain:
 - Out-of-vocabulary words
 - Domain mismatch between train and test data
 - Maintaining context over longer text
 - Low-resource language pairs

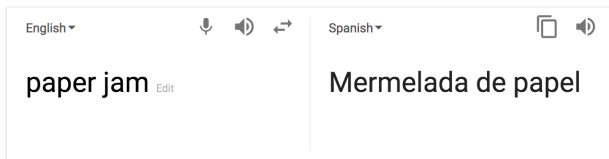
Further reading: *"Has AI surpassed humans at translation? Not even close!"*
https://www.skynettoday.com/editorials/state_of_nmt

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So is Machine Translation solved?

- **Nope!**
- Using **common sense** is still hard



[Open in Google Translate](#)

[Feedback](#)

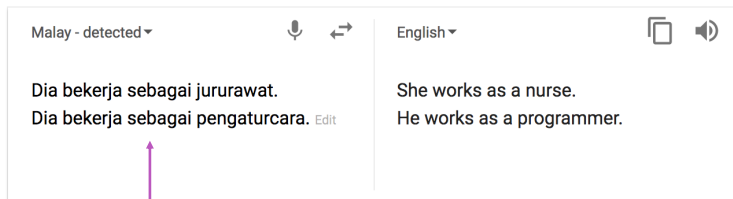


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So is Machine Translation solved?

- **Nope!**
- NMT picks up **biases** in training data



Didn't specify gender

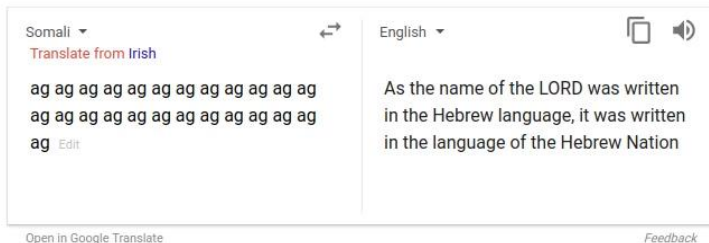
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Source: <https://hackernoon.com/bias-sexist-or-this-is-the-way-it-should-be-ce1f7c8c683c>

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So is Machine Translation solved?

- Nope!
- Uninterpretable systems do strange things



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Picture source: https://www.vice.com/en_uk/article/j5npeg/why-is-google-translate-spitting-out-sinister-religious-prophecies

Explanation: <https://www.skynettoday.com/briefs/google-nmt-prophecies>

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NMT research continues

NMT is the **flagship task** for NLP Deep Learning

- NMT research has **pioneered** many of the recent **innovations** of NLP Deep Learning
- In **2019**: NMT research continues to **thrive**
 - Researchers have found **many, many improvements** to the “vanilla” seq2seq NMT system we’ve presented today
 - But **one improvement** is so integral that it is the new vanilla...

ATTENTION

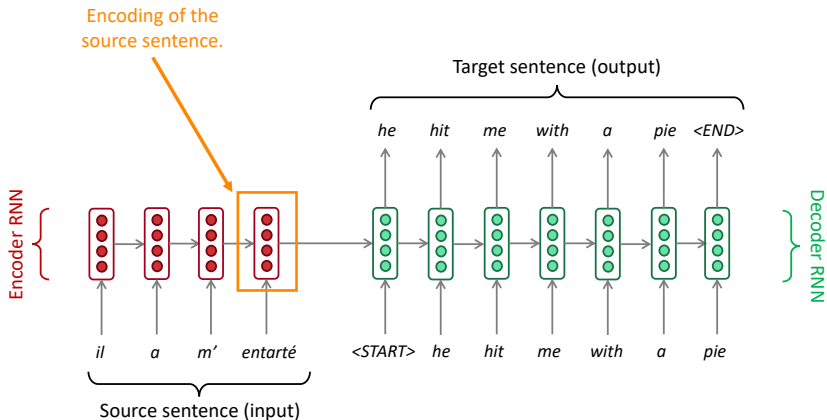
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Content

- 1 Machine Translation (MT)
- 2 Machine translation evaluation
- 3 Statistical machine translation (SMT)
- 4 Neural machine translation (NMT) based on RNNs
- 5 Neural machine translation (NMT) with attentions

Sequence-to-sequence: the bottleneck problem

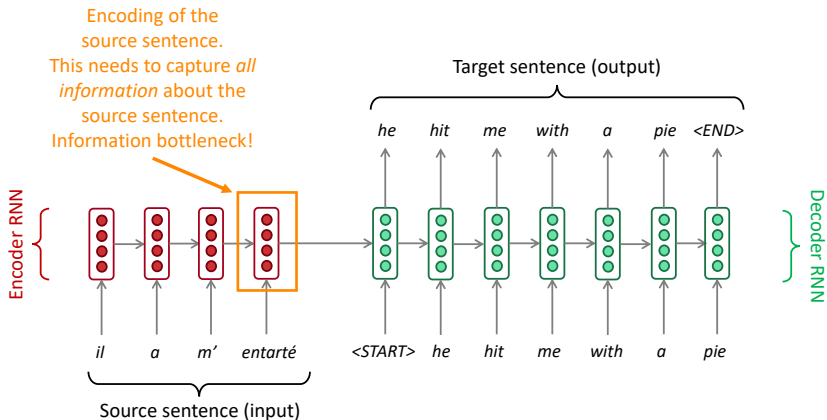


Problems with this architecture?

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Sequence-to-sequence: the bottleneck problem



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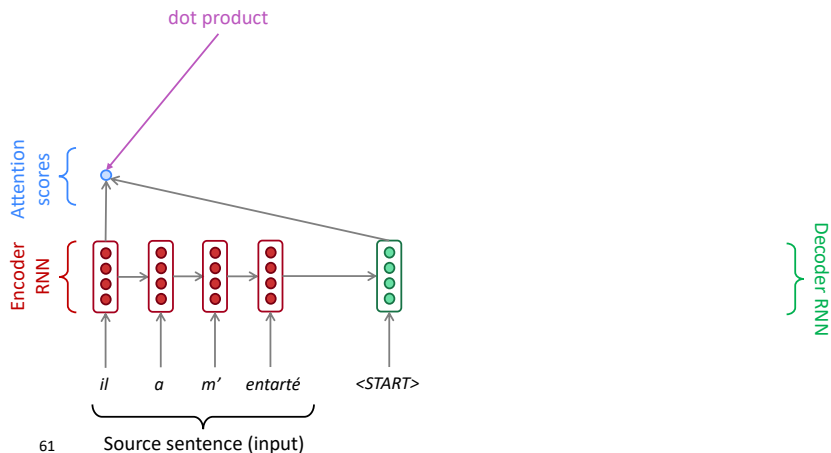
Attention

- **Attention** provides a solution to the bottleneck problem.
- Core idea: on each step of the decoder, use *direct connection to the encoder* to *focus on a particular part* of the source sequence



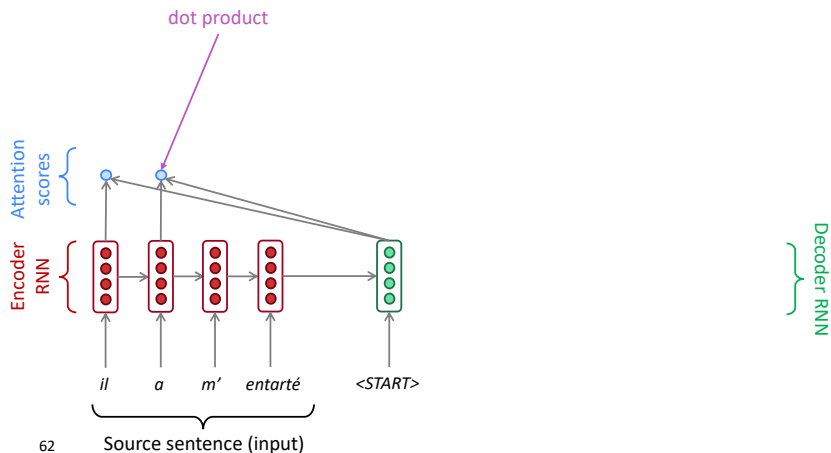
- First we will show via diagram (no equations), then we will show with equations

Sequence-to-sequence with attention



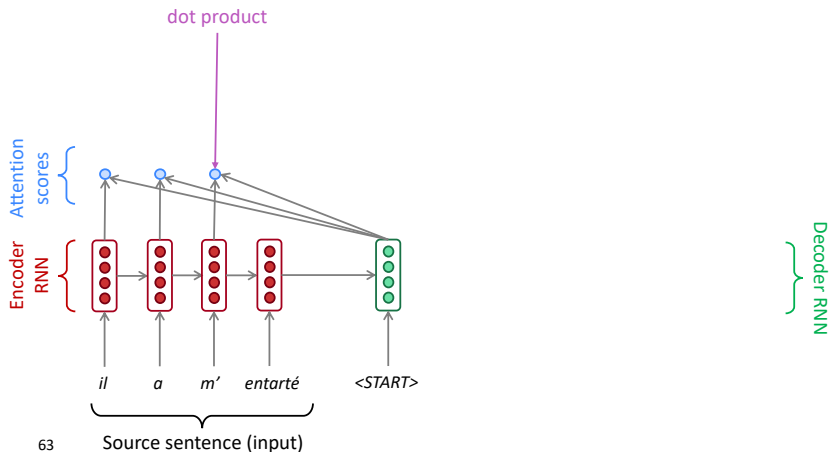
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Sequence-to-sequence with attention



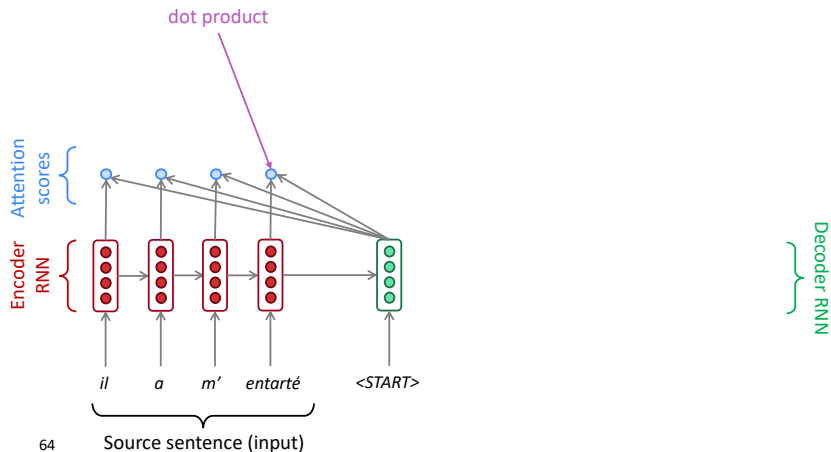
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Sequence-to-sequence with attention



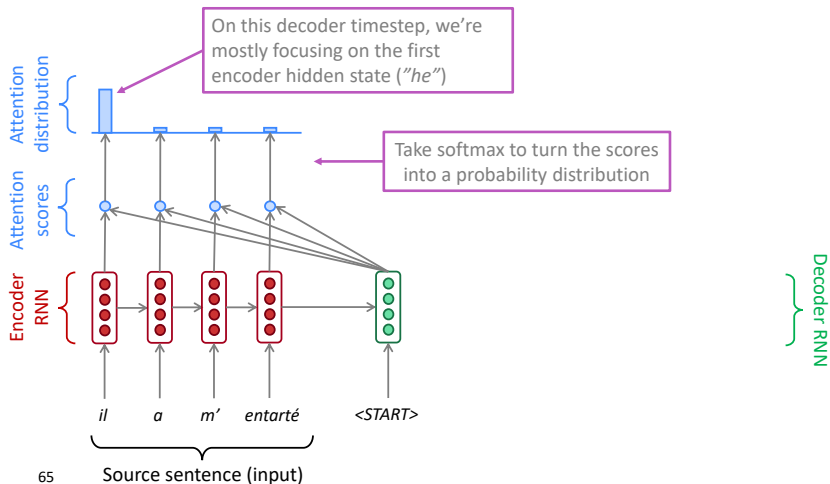
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Sequence-to-sequence with attention



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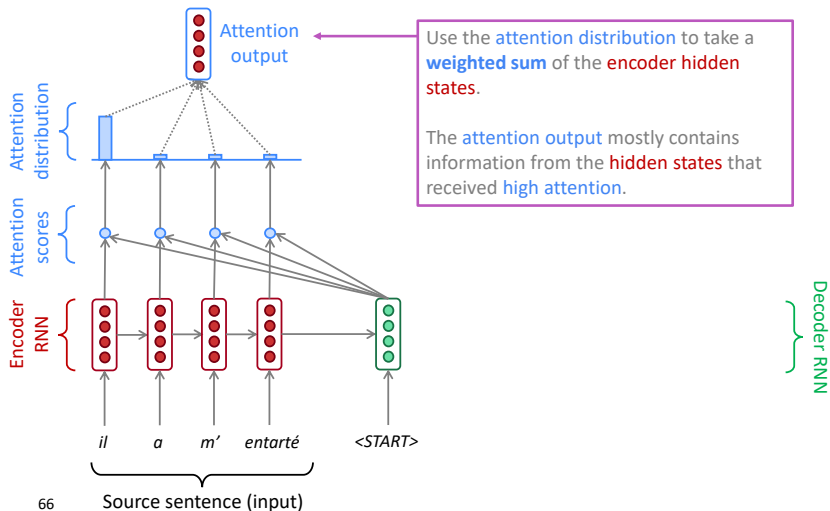
Sequence-to-sequence with attention



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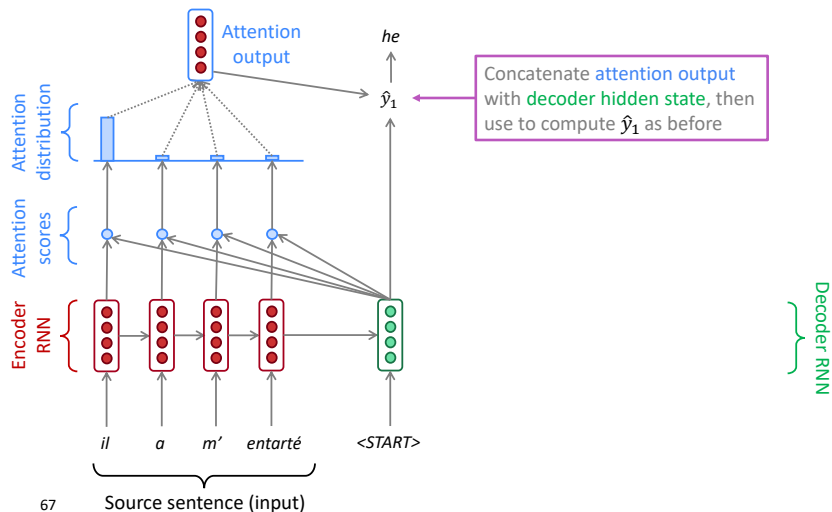
Sequence-to-sequence with attention



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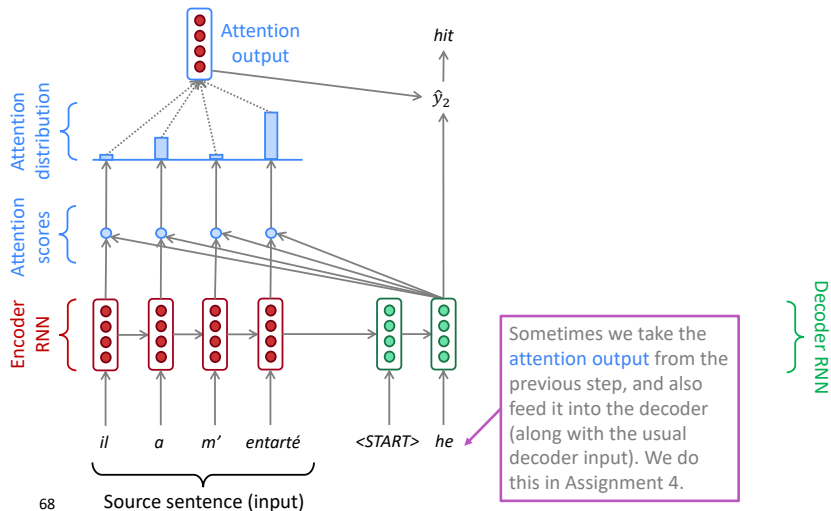
Christopher Manning, Natural Language Processing with Deep Learning, 2019 (slides)

Sequence-to-sequence with attention



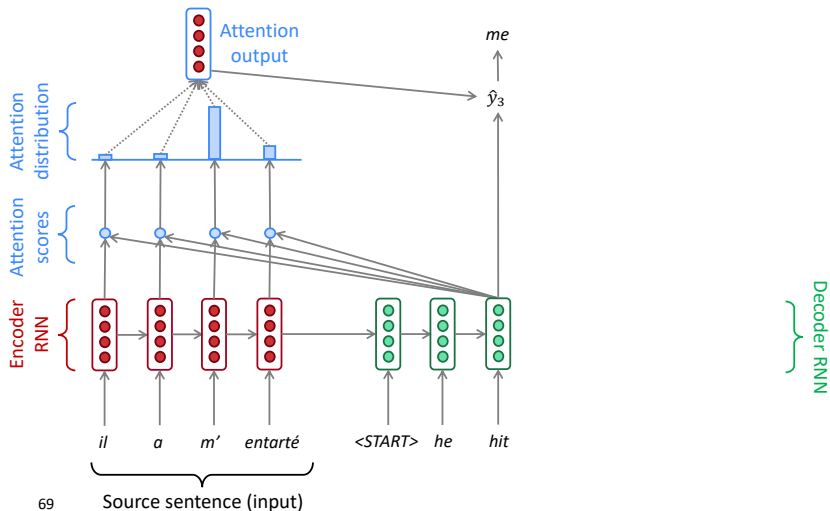
Christopher Manning, Natural Language Processing with Deep Learning, 2019 (slides)

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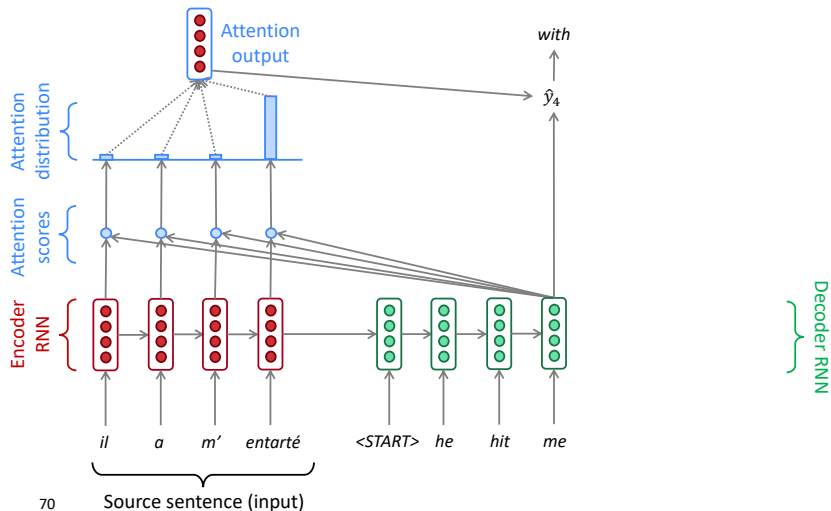
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Sequence-to-sequence with attention



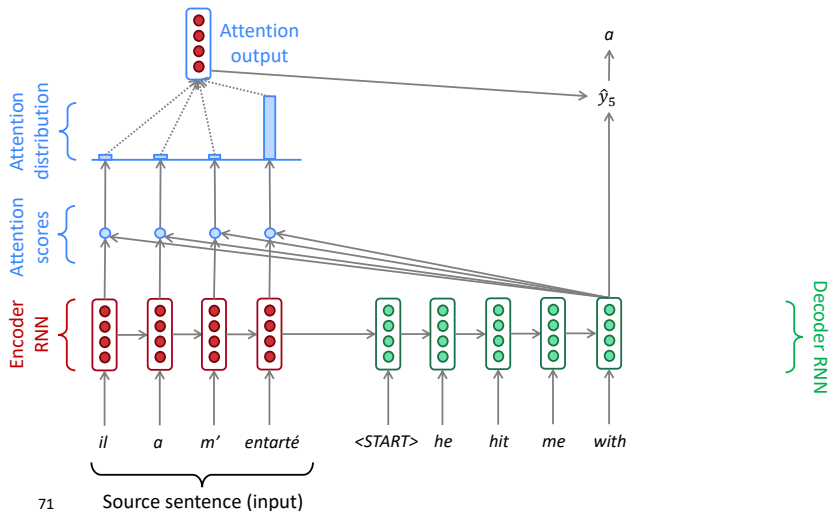
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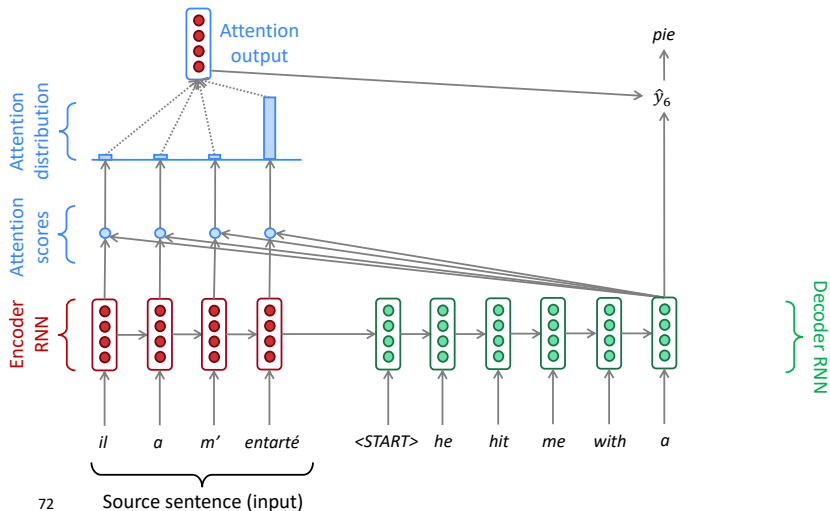
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Sequence-to-sequence with attention



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Sequence-to-sequence with attention



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Attention: in equations

- We have encoder hidden states $h_1, \dots, h_N \in \mathbb{R}^h$
- On timestep t , we have decoder hidden state $s_t \in \mathbb{R}^h$
- We get the attention scores e^t for this step:

$$e^t = [s_t^T h_1, \dots, s_t^T h_N] \in \mathbb{R}^N$$

- We take softmax to get the attention distribution α^t for this step (this is a probability distribution and sums to 1)

$$\alpha^t = \text{softmax}(e^t) \in \mathbb{R}^N$$

- We use α^t to take a weighted sum of the encoder hidden states to get the attention output a_t

$$a_t = \sum_{i=1}^N \alpha_i^t h_i \in \mathbb{R}^h$$

- Finally we concatenate the attention output a_t with the decoder hidden state s_t and proceed as in the non-attention seq2seq model

$$[a_t; s_t] \in \mathbb{R}^{2h}$$

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Attention is great

- Attention significantly **improves NMT performance**
 - It's very useful to allow decoder to focus on certain parts of the source
- Attention **solves the bottleneck problem**
 - Attention allows decoder to look directly at source; bypass bottleneck
- Attention **helps with vanishing gradient problem**
 - Provides shortcut to faraway states
- Attention provides **some interpretability**
 - By inspecting attention distribution, we can see what the decoder was focusing on
 - We get (soft) **alignment for free!**
 - This is cool because we never explicitly trained an alignment system
 - The network just learned alignment by itself

	he	hit	me	with	a	pie
il	black					
a		gray				
m'			black			
entarté		black	gray	black	black	black

Attention is a *general* Deep Learning technique

- We've seen that attention is a great way to improve the sequence-to-sequence model for Machine Translation.
 - However: You can use attention in *many architectures* (not just seq2seq) and *many tasks* (not just MT)
- More general definition of attention:
 - Given a set of vector *values*, and a vector *query*, attention is a technique to compute a weighted sum of the values, dependent on the query.
- We sometimes say that the *query attends to the values*.
 - For example, in the seq2seq + attention model, each decoder hidden state (query) *attends to* all the encoder hidden states (values).

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Attention is a *general* Deep Learning technique

More general definition of attention:

Given a set of vector *values*, and a vector *query*, attention is a technique to compute a weighted sum of the values, dependent on the query.

Intuition:

- The weighted sum is a *selective summary* of the information contained in the values, where the query determines which values to focus on.
- Attention is a way to obtain a *fixed-size representation of an arbitrary set of representations* (the values), dependent on some other representation (the query).

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